Local classification: one sensing layer

ONE piezo sensor + ONE ESP32-WROOM32 + ONE RFM95W link. Temporal windows are not extra sensing layers.

1 WAKE AND ACQUIRE

BOOT / INITIALIZE

Load the matched deployment profile
Feature constants, versions and thresholds
Restore RTC schedule, daily state and queue

Wake from RTC or pending retry
Check current time and unsent state
Measurement cadence is configurable

Measurement due?
No → C
Yes

Power one sensing chain
One piezo disc + MCP6002 conditioning
Enable ESP32 ADC / DMA

Acquire one streamed recording
Example: 20.5 s at 8 kHz mono
Stream with bounded / reused buffers
Keep the acquisition timestamp

Acquisition valid?
ADC / clipping / data checks
Yes → A
No

Record an invalid measurement
Increment quality / acquisition counters
Do not classify an invalid capture as healthy
→ C
Report status

2 PROCESS THE WINDOWS

A

Select the temporal windows
10 s windows, 5 s stride + final tail
20.5 s example: 0, 5, 10, 10.5 s
Reset this recording's maximum score

Filter and measure signal quality
Bandpass: 100-2500 Hz
100 ms energy / burst statistics
Energy variation is not an RPW decision

Extract and normalize MFCCs
30 frames x 13 coefficients = 390 values
Apply frozen training mean / scale
Match the validated frontend exactly

Run the local INT8 residual CNN
Quantize inputs with export scale / zero
Run the saved 55,864-byte model
Dequantize the output logit

Update the recording RPW score
Apply sigmoid to the window logit
Record score = maximum window score
Release / reuse temporary tensors

More windows?
Yes: next window
No

Apply the saved recording threshold
RPW or healthy from the maximum score
Update valid / RPW / healthy counters
Retain max score and quality statistics

C

3 REPORT AND DELIVER

C

Persist daily state
Measurement and validity / RPW counters
Max score, battery and quality / fault flags
Retain unacknowledged prior days

Report or retry due?
At least one report each day
Yes
No

Build the daily result summary
Proposed 24-byte application body
ID / day, counts, scores, battery / status
Report even when no measurement is valid

Queue the packet before radio TX
Use sequence and schema / model version
Keep totals until acknowledgement
Transport / integrity metadata is extra

Transmit using RFM95W LoRa
Use the configured channel / SF / BW
Apply the planned TX slot / backoff
Gateway relays; queued data survives failure

Application ACK received?
No
Yes

Delivery pending
Bounded retries
Backoff / retry later
Persist until ACK
Mark report overdue

Commit acknowledged state
Clear acknowledged items / totals only
Queue pending: schedule immediate TX wake
Else sleep until next sample / retry

Next scheduled wake

EXTERNAL ENDPOINT ONLY - gateway relay and result storage
Gateway: validate / buffer the LoRa packet and forward over Wi-Fi or cellular.
Server: deduplicate by node / day / sequence, store daily results and update dashboard / RPW alerts.
No server acoustic classifier is required for this route; the node already made the recording-level decisions.

The 24-byte daily summary is a proposed body, not a measured complete radio frame. Reporting schedule and ACK/retry protocol require firmware implementation.

Local model is an INT8 deployment candidate. Feature parity, contiguous RAM, inference timing and energy must be measured on the actual ESP32.

Daily reporting remains required even after rejected captures. The energy-variation proxy is a quality statistic, not proof of RPW activity.

Server-side classification: one sensing layer

ONE piezo sensor + ONE ESP32-WROOM32 + ONE RFM95W link. Temporal windows are not extra sensing layers.

1 WAKE AND ACQUIRE

BOOT / INITIALIZE

Load the matched deployment profile
Feature constants, versions and thresholds
Restore RTC schedule, daily state and queue

Wake from RTC or pending retry
Check current time and unsent state
Measurement cadence is configurable

Measurement due?
No → C
Yes

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Acquire one streamed recording
Example: 20.5 s at 8 kHz mono
Stream with bounded / reused buffers
Keep the acquisition timestamp

Acquisition valid?
ADC / clipping / data checks
Yes → A
No

Record an invalid measurement
Increment quality / acquisition counters
Do not classify an invalid capture as healthy
→ C
Report status

2 PROCESS THE WINDOWS

A

Select the temporal windows
10 s windows, 5 s stride + final tail
20.5 s example: 0, 5, 10, 10.5 s
Initialize ordered window-vector list

Filter and measure signal quality
Bandpass: 100-2500 Hz
100 ms energy / burst statistics
Energy variation is not an RPW decision

Extract and normalize MFCCs
30 frames x 13 coefficients = 390 values
Apply frozen training mean / scale
Match the validated frontend exactly

Project into PCA16 coordinates
Subtract the frozen PCA training mean
Apply the learned 16 x 390 projection
No classifier runs on this sensing node

Encode one 16-byte window vector
Round / clamp using training-fitted ranges
16 components, one byte per component
Retain profile ID and window order

More windows?
Yes: next window
No

Losslessly pack the window vectors
First vector + delta / Rice codes + escapes
Keep the shortest encoding or raw fallback
No additional PCA precision is discarded

C

3 REPORT AND DELIVER

C

Persist features and health state
Keep each complete measurement record
Node / record / window / profile identifiers
No RPW label is generated here

Transmit due?
Sample / daily / retry deadline
Yes
No

Build the feature / status report
Drain retained records at least daily
Multiple packets may be required
Send heartbeat even with no valid capture

Feature block fits budget?
64 B feature-block target
No → **If oversized**
Fragment the record
Keep index + total
Preserve all windows
Do not silently drop
Yes

Transmit using RFM95W LoRa
Use the configured channel / SF / BW
Apply the planned TX slot / backoff
Gateway relays; queued data survives failure

Application ACK received?
No → **Delivery pending**
Bounded retries
Backoff / retry later
Persist until ACK
Mark report overdue
Yes

Commit acknowledged state
Clear acknowledged items / totals only
Queue pending: schedule immediate TX wake
Else sleep until next sample / retry

Next scheduled wake

EXTERNAL ENDPOINT ONLY - gateway relay and server classification
Gateway: validate / buffer / forward over Wi-Fi or cellular; no RPW inference at the gateway.
Server: check profile/version, reassemble fragments, unpack deltas, dequantize PCA16 and apply inverse PCA.
Reshape to normalized 30 x 13 MFCCs (do not normalize twice); classify windows, take maximum score, apply saved threshold, store / alert.

Current evaluated classifier route: four windows for a 20.5 s recording. Five-window delta packing is a payload experiment, not a revalidated classifier.

64 B refers to a feature-block target. Node/time identifiers, fragment headers, integrity/security and radio framing consume additional bytes.

Daily reporting remains required even after rejected captures. The energy-variation proxy is a quality statistic, not proof of RPW activity.